WORKSHEET — UNIT I

Foundations of Quantum Computing: Qubits, Superposition and Entanglement

Course: Fundamentals of Quantum Computing

Unit: I (5 Hrs / 300 min)

Topics: Course Introduction; Mathematical Toolkit (Linear Algebra); Classical Bits vs Qubits; Superposition; The Bloch Sphere; Measurement Postulates; Multi-Qubit Systems and Entanglement; Hands-On with a Quantum Simulator

🎯 Section A — Learning Objectives

1. Explain why classical computers struggle with cryptography, chemistry and optimization, and separate the hype from what quantum computers can do today.

2. Use the linear-algebra toolkit: complex numbers, vectors, inner products, unitary and Hermitian matrices, the tensor product and Hilbert space.

3. Describe a qubit in Dirac notation, compare it with a classical bit, and explain why n qubits carry 2^n amplitudes.

4. Explain superposition, probability amplitudes, normalization, global versus relative phase and interference.

5. Map single-qubit states onto the Bloch sphere using the angles (θ, φ) and distinguish pure from mixed states.

6. Apply the Born rule, measurement collapse and the no-cloning theorem; build multi-qubit states, identify Bell states and run a first Qiskit simulation.

🔑 Section B — Key Concepts

No.	Concept	Short Explanation
1	**Why Quantum Computing**	Classical machines need exponential resources for problems such as factoring, molecular simulation and some optimization tasks; quantum computers use quantum effects to attack them.
2	**Complex Numbers and Vectors**	A complex number a + bi has a modulus and a phase (Euler form r·e^{iθ}); quantum states are complex vectors in a vector space.
3	**Inner Product and Norm**	The inner product ⟨ψ|φ⟩ measures overlap between vectors; the norm √⟨ψ|ψ⟩ is the length, and normalization makes it equal to 1.
4	**Unitary and Hermitian Matrices**	A unitary matrix satisfies U†U = I and preserves length; a Hermitian matrix satisfies H† = H and has real eigenvalues.
5	**Tensor Product and Hilbert Space**	The tensor (Kronecker) product combines separate systems into one larger system; a Hilbert space is the complex inner-product space in which quantum states live.
6	**Qubit and Dirac Notation**	A qubit is a unit vector α|0⟩ + β|1⟩ in a 2-dimensional complex space; |0⟩ and |1⟩ form the computational basis, and |+⟩, |−⟩ form another basis.
7	**Superposition and Phase**	A qubit can be a linear combination of basis states with amplitudes α and β, where $\|\alpha\|^2 + \|\beta\|^2 = 1$; relative phase, not global phase, affects interference.
8	**Bloch Sphere**	Two real angles (θ, φ) describe any pure qubit state as a point on a sphere; gates appear as rotations of that sphere.
9	**Measurement and No-Cloning**	The Born rule gives probability = |amplitude|²; measurement collapses the state irreversibly, and an unknown state cannot be copied.
10	**Entanglement**	Entangled multi-qubit states cannot be written as a product of single-qubit states; the Bell states are the standard two-qubit examples.

📘 Section C — Important Terminology

Keyword	Meaning
Modulus	The size of a complex number, $\|a + bi\| = \sqrt{(a^2 + b^2)}$.
Euler Form	Writing a complex number as $r \cdot e^{i\theta}$, with modulus r and phase angle θ.
Inner Product	A complex-valued product $\langle\psi\|\varphi\rangle$ of two vectors that measures their overlap.
Norm	The length of a vector, $\sqrt{\langle\psi\|\psi\rangle}$; a valid quantum state has norm 1.
Dagger (†)	The conjugate transpose of a matrix or vector: transpose it and take the complex conjugate of every entry.
Unitary Matrix	A square matrix with $U^\dagger U = I$; it preserves norms and represents a reversible quantum operation.
Hermitian Matrix	A matrix equal to its own conjugate transpose ($H^\dagger = H$); its eigenvalues are real.
Eigenvector / Eigenvalue	A vector v and number λ with $Av = \lambda v$; the matrix only stretches v by λ.
Tensor Product	The Kronecker product ⊗ that joins the state spaces of separate systems.
Hilbert Space	A complex vector space with an inner product, used as the "stage" for quantum states.
Ket / Bra	$\|\psi\rangle$ is a column-vector state (ket); $\langle\psi\|$ is its conjugate-transpose row vector (bra).
Computational Basis	The basis $\{\|0\rangle, \|1\rangle\}$ used for reading out qubits.
Probability Amplitude	The complex coefficient of a basis state; its squared modulus is a probability.
Global Phase	A common factor $e^{i\gamma}$ on the whole state; it has no observable effect.
Relative Phase	The phase difference between components of a superposition; it changes interference.
Interference	Amplitudes adding (constructive) or cancelling (destructive) before measurement.
Bloch Sphere	A unit sphere whose surface points represent the pure states of one qubit.
Born Rule	The probability of an outcome equals the squared modulus of its amplitude.
Shot	One complete run of a circuit ending in a measurement; many shots build a histogram.
No-Cloning Theorem	No operation can copy an arbitrary unknown quantum state.
Bell State	One of four maximally entangled two-qubit states, e.g. $(\|00\rangle + \|11\rangle)/\sqrt{2}$.
Qiskit	An open-source Python toolkit for building, simulating and running quantum circuits, including on the IBM Quantum cloud.

✅ Section D — Multiple-Choice Questions [Easy–Medium]

1. Which set lists three areas used to motivate quantum computing?

- **A.** Cryptography, chemistry and optimization
- **B.** Word processing, web browsing and email
- **C.** Printing, scanning and copying
- **D.** Video editing, gaming and streaming

2. Which type of matrix satisfies $U^\dagger U = I$ and preserves the length of a vector?

- **A.** Hermitian matrix
- **B.** Transpose matrix
- **C.** Unitary matrix
- **D.** Diagonal matrix

3. Mathematically, a single qubit is a unit vector in which space?

A. A 1-dimensional real space
B. A 2-dimensional complex space
C. A 4-dimensional real space
D. A 2-dimensional real space

4. How many complex amplitudes are needed to describe an n-qubit state?

A. n
B. 2n
C. n^2
D. 2^n

5. Which state is correctly normalized?

A. $(1/2)|0\rangle + (1/2)|1\rangle$
B. $(1/\sqrt{2})|0\rangle + (1/\sqrt{2})|1\rangle$
C. $|0\rangle + |1\rangle$
D. $2|0\rangle - |1\rangle$

6. Which phase has no observable effect on measurement outcomes?

A. Relative phase
B. Both global and relative phase
C. Global phase
D. Neither global nor relative phase

7. On the standard Bloch sphere, which state sits at the north pole?

A. $|0\rangle$
B. $|1\rangle$
C. $|+\rangle$
D. $|-i\rangle$

8. A qubit is in the state $(\sqrt{3}/2)|0\rangle + (1/2)|1\rangle$. What is the probability of measuring 0?

A. 1/2
B. 1/4
C. $\sqrt{3}/2$
D. 3/4

9. What does the no-cloning theorem state?

A. Every qubit can be copied using a CNOT gate
B. An arbitrary unknown quantum state cannot be copied exactly
C. Cloning needs at least two qubits and no more
D. Cloning works only for states on the Bloch equator

10. Which of the following is a Bell state?

A. $|00\rangle$
B. $[(|0\rangle + |1\rangle)/\sqrt{2}] \otimes [(|0\rangle + |1\rangle)/\sqrt{2}]$
C. $(|00\rangle + |11\rangle)/\sqrt{2}$
D. $|01\rangle$

✏️ Section E — Fill in the Blanks [Easy]

1. The conjugate transpose of a matrix A is written A† and is called the _____ of A.

2. In Dirac notation, the column-vector state $|\psi\rangle$ is called a _____.

3. The probability of an outcome equals the squared modulus of its amplitude; this is the _____ rule.

4. A _____ phase multiplies the whole state by e^{iγ} and cannot be detected by any measurement.

5. Two qubits whose joint state cannot be written as a tensor product of single-qubit states are said to be _____.

☑️ Section F — True or False [Easy]

1. A Hermitian matrix equals its own conjugate transpose, and its eigenvalues are real numbers. (True / False)

2. A qubit in superposition is simply either 0 or 1, and we just do not know which. (True / False)

3. Measuring $(|0\rangle + |1\rangle)/\sqrt{2}$ in the computational basis gives 0 or 1, each with probability 1/2. (True / False)

4. Mixed states lie on the surface of the Bloch sphere. (True / False)

5. The Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$ is an entangled state. (True / False)

🔗 Section G — Match the Following [Medium]

Column A	Column B (shuffled)
1. Ket $\|\psi\rangle$	a. Maximally entangled two-qubit state such as $(\|00\rangle + \|11\rangle)/\sqrt{2}$
2. Unitary matrix	b. Geometric picture of a pure qubit state using two angles
3. Hilbert space	c. Rule that turns the squared modulus of an amplitude into a probability
4. Born rule	d. Column-vector notation for a quantum state
5. Bloch sphere	e. Complex inner-product space in which quantum states live
6. No-cloning theorem	f. Matrix with U†U = I that represents a reversible operation
7. Bell state	g. Impossibility of copying an arbitrary unknown quantum state

📝 Section H — Short-Answer Questions [Medium]

1. Name three areas where classical computers hit a wall and say why quantum computing is considered promising for them.

2. How does a qubit differ from a classical bit? Include the condition on its amplitudes.

3. Distinguish global phase from relative phase, with one example of each.

4. State the Born rule and use it to find the probabilities for $|\psi\rangle = (1/\sqrt{3})|0\rangle + \sqrt{(2/3)}|1\rangle$.

5. State the no-cloning theorem and explain why it matters in practice.

📄 Section I — Descriptive Questions [Medium–Advanced]

1. Explain the linear-algebra toolkit for quantum computing — complex numbers, inner product and norm, unitary and Hermitian matrices, the tensor product and Hilbert space — and say how each is used.

2. Describe the qubit and superposition: Dirac notation, computational and alternative bases, amplitudes, the normalization condition, global versus relative phase, interference, and why "a qubit is both 0 and 1 at once" is a misleading statement.

3. Explain the Bloch sphere: the angles (θ, φ), the positions of $|0\rangle$, $|1\rangle$, $|+\rangle$, $|-\rangle$, $|+i\rangle$ and $|-i\rangle$, rotations as operations, and pure versus mixed states. Convert a given state to Bloch coordinates.

4. Discuss measurement and entanglement: the Born rule, collapse, measuring in other bases, shots and histograms, the no-cloning theorem, product versus entangled states, the four Bell states, GHZ and W states, the EPR paradox and Bell inequality, and entanglement as a resource.

💻 Section J — Practical / Application-Based Questions [Advanced]

Practical 1

Problem: Write a Python function that turns raw amplitudes into measurement probabilities.

Task:

1. Define qubit_probabilities(alpha, beta), which normalizes the state first (divide by $|\alpha|^2 + |\beta|^2$).

2. Return the probabilities of measuring 0 and 1 as a tuple, rounded to 2 decimal places.

3. Test it with (3, 4j), (1, 1) and (0, 2j) and print each result.

Expected Output: *Three lines: (0.36, 0.64), (0.5, 0.5), (0.0, 1.0).*

Practical 2

Problem: Compute Bloch-sphere coordinates for a single-qubit state.

Task:

1. Define bloch_vector(alpha, beta) using $x = 2 \cdot \mathrm{Re}(\alpha^* \beta)$, $y = 2 \cdot \mathrm{Im}(\alpha^* \beta)$ and $z = |\alpha|^2 - |\beta|^2$.

2. Round each coordinate to 2 decimal places and return (x, y, z).

3. Test it with $|0\rangle$, $|+\rangle = (1/\sqrt{2}, 1/\sqrt{2})$ and $|+i\rangle = (1/\sqrt{2}, i/\sqrt{2})$ and print each result.

Expected Output: *Three lines: (0.0, 0.0, 1.0), (1.0, 0.0, 0.0), (0.0, 1.0, 0.0).*

Practical 3

Problem: Build two-qubit states with the tensor product and test whether they are entangled.

Task:

1. Write kron(u, v) that returns the Kronecker product of two 2-element lists as a 4-element list.

2. Write is_separable(state) for a state [a, b, c, d], returning True when $|ad - bc|$ is below 1e-9.

3. Print kron([1, 0], [0, 1]), then test $|+\rangle \otimes |0\rangle$ and the Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$ with is_separable.

Expected Output: *Three lines: [0, 1, 0, 0], True, False.*

🗺️ Section K — Concept-Mapping Activity

QUANTUM FOUNDATIONS

Complete the map below: fill in the missing sub-topic name or example for each branch.

Branch 1	Branch 2	Branch 3	Branch 4
Mathematical Toolkit	**Qubits and Superposition**	**Bloch Sphere and Measurement**	**Entanglement**
Example: _____	Example: _____	Example: _____	Example: _____

(Answer key — example values: unitary matrices, tensor product, Hilbert space; Dirac notation, amplitudes, interference; Born rule, collapse, no-cloning; Bell states, GHZ state, teleportation)

🔬 Section L — Case Study / Real-World Activity

Scenario

A lab group uses Qiskit on a simulator to study one qubit. They prepare the state $(|0\rangle + |1\rangle)/\sqrt{2}$ and run 1024 shots; the histogram shows 508 results of "0" and 516 results of "1". They then prepare the Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$ and measure both qubits: only "00" and "11" ever appear, in roughly equal numbers. A group member suggests copying the qubit first so that the same state can be measured again and again.

Questions

1. What probabilities does the Born rule predict for the single-qubit state, and why are the counts 508 and 516 rather than exactly 512 each?

2. In the Bell-pair experiment, the first qubit is measured and gives 1. What is the state of the second qubit, and what does this show about entanglement?

3. Why can the group not simply copy the qubit to repeat the measurement? What must they do instead to get a histogram?

4. Name one Qiskit visualizer from the hands-on session that would show the state before measurement, and one physical qubit technology the experiment could run on.

Expected Learning Outcome

Students apply the Born rule, measurement collapse, the no-cloning theorem and entanglement to interpret the results of a simple quantum-simulator experiment.

📊 Section M — Self-Assessment

☐ **Yes** ☐ **Partly** ☐ **No** I can explain the major concepts of this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can define the important terminology from this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can solve problems/exercises from this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can apply the concepts of this unit to a new situation.

💭 Section N — Student Reflection

What did I learn in this unit?

What was difficult for me?

What do I need to practice more?

One important concept I learned:

🔑 Section O — Instructor Answer Key

(For instructor use — covers all objective and subjective items in this worksheet.)

MCQ Answers

- 1. A — Cryptography (factoring), chemistry (molecular simulation) and optimization are the standard motivating areas.
- 2. C — A unitary matrix satisfies $U^\dagger U = I$, so it preserves length; a Hermitian matrix satisfies $H^\dagger = H$ instead.
- 3. B — A qubit is a unit vector in a 2-dimensional complex vector space.
- 4. D — Each of the 2^n basis states has its own amplitude, so the state space grows exponentially.

- 5. B — $(1/\sqrt{2})^2 + (1/\sqrt{2})^2 = 1/2 + 1/2 = 1$; the other options give 1/2, 2 and 5.
- 6. C — A global phase multiplies the whole state and cancels in every probability; relative phase does not cancel.
- 7. A — With the standard convention, $|0\rangle$ is the north pole and $|1\rangle$ the south pole.
- 8. D — $P(0) = |\sqrt{3}/2|^2 = 3/4$, and $P(1) = 1/4$.
- 9. B — No quantum operation can make an independent copy of an arbitrary unknown state.
- 10. C — $(|00\rangle + |11\rangle)/\sqrt{2}$ is a Bell state; option B is a product state and A and D are basis states.

Fill in the Blanks — Answers

- 1. adjoint (Hermitian conjugate)
- 2. ket
- 3. Born
- 4. global
- 5. entangled

True/False — Answers

- 1. True
- 2. False
- 3. True
- 4. False
- 5. True

Match the Following — Answer Key

1 → d 2 → f 3 → e 4 → c 5 → b 6 → g 7 → a

Short Answer — Key Points

- 1. Cryptography (factoring large integers), chemistry (simulating molecules needs memory that grows exponentially) and optimization (huge search spaces). Quantum states can encode many amplitudes at once and use interference, so some of these problems may be solved faster; the gains depend on the problem.
- 2. A bit is deterministically 0 or 1. A qubit is a unit vector $\alpha|0\rangle + \beta|1\rangle$ with complex amplitudes and $|\alpha|^2 + |\beta|^2 = 1$; n qubits need 2^n amplitudes, while n bits hold only n values.
- 3. Global phase is a common factor e^{iγ} on the whole state, e.g. $|\psi\rangle$ and $-|\psi\rangle$, and is unobservable. Relative phase is the phase between components, e.g. $(|0\rangle + |1\rangle)/\sqrt{2}$ versus $(|0\rangle - |1\rangle)/\sqrt{2}$, and changes interference and measurements in other bases.
- 4. Born rule: probability = |amplitude|². Here $P(0) = 1/3$ and $P(1) = 2/3$, which add to 1.
- 5. No arbitrary unknown state $|\psi\rangle$ can be copied to $|\psi\rangle|\psi\rangle$. It prevents simple backup copies and repetition-based error correction, and it underlies the security of quantum communication; to gather statistics, the state must be prepared again for each shot.

Descriptive Questions — Key Points

- 1. Complex numbers give amplitudes and phase (modulus, Euler form); the inner product and norm give overlap, length and normalization; the dagger gives conjugate transposes; unitary matrices model reversible operations and Hermitian matrices have real eigenvalues (observables); the tensor product builds multi-qubit systems; the Hilbert space is where all states live.
- 2. Cover $|0\rangle$, $|1\rangle$, $\langle\psi|$, $\langle\psi|\varphi\rangle$, the computational basis versus $|+\rangle$ and $|-\rangle$, $\alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$, global versus relative phase, constructive and destructive interference, and the equal superposition example. Misconception: before measurement the qubit has a definite quantum state with amplitudes, and measurement yields only one outcome with the stated probability.
- 3. A pure state is $\cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle$. $|0\rangle$ and $|1\rangle$ are the poles, $|+\rangle$ and $|-\rangle$ lie on the x-axis, $|+i\rangle$ and $|-i\rangle$ on the y-axis. Gates are rotations. Pure states lie on the surface, mixed states inside. Example: $(1/\sqrt{2}, i/\sqrt{2})$ gives (x, y, z) = (0, 1, 0).
- 4. Born rule and collapse (irreversible); measuring in other bases; shots and histograms; no-cloning; product versus entangled states (ad − bc test); Φ^+, Φ^-, Ψ^+, Ψ^-; GHZ and W states; correlations when one half is measured; EPR paradox and Bell inequality conceptually; entanglement as a resource for teleportation and superdense coding.

Practical Activities — Expected Outputs

- 1. Sample logic: def qubit_probabilities(alpha, beta): total = abs(alpha)**2 + abs(beta)**2; return round(abs(alpha)**2/total, 2), round(abs(beta)**2/total, 2). The tests print (0.36, 0.64), (0.5, 0.5) and (0.0, 1.0).
- 2. Sample logic: a = complex(alpha); b = complex(beta); p = a.conjugate() * b; x = 2*p.real; y = 2*p.imag; z = abs(a)**2 − abs(b)**2, then round each value. The tests give (0.0, 0.0, 1.0), (1.0, 0.0, 0.0) and (0.0, 1.0, 0.0).
- 3. Sample logic: kron(u, v) = [u[0]*v[0], u[0]*v[1], u[1]*v[0], u[1]*v[1]]; is_separable returns abs(a*d − b*c) < 1e-9. The output is [0, 1, 0, 0], True (ad − bc = 0) and False (ad − bc = 1/2).

Concept Map — Sample Answers

- Mathematical Toolkit: complex numbers, inner product, unitary and Hermitian matrices, tensor product, Hilbert space. Qubits and Superposition: ket |0⟩, |+⟩ basis, amplitudes, global and relative phase, interference. Bloch Sphere and Measurement: angles θ and φ, poles and equator, Born rule, collapse, no-cloning. Entanglement: Bell states, GHZ and W states, EPR and Bell inequality, teleportation.

Case Study — Key Points

- 1. P(0) = P(1) = 1/2. Each shot is an independent random outcome, so counts fluctuate around 512; they approach 1/2 as the number of shots grows.
- 2. The second qubit collapses to |1⟩ with certainty. The two outcomes are perfectly correlated, which a product state cannot reproduce.
- 3. The no-cloning theorem forbids copying an unknown state. The group must re-prepare the state and measure it once per shot.
- 4. Visualizers: Qiskit state-vector or Bloch-sphere plot. Qubit technologies: superconducting, trapped-ion, photonic or spin qubits.